

Legal & Contract Clause Review System

This document describes the functional and non-functional requirements of the Legal & Contract Clause Review System. The system is intended to assist legal in reviewing business contracts by automatically extracting clauses, comparing them against company standards, and highlighting potential risks.

1. Scope

The system analyzes contracts such as NDAs, MSAs, SOWs.. It supports multiple document formats and provides structured outputs to reduce manual review effort while ensuring legal judgment remains with humans.

Definitions, Acronyms, and Abbreviations

Term	Meaning
NDA	Non-Disclosure Agreement
MSA	Master Service Agreement
RA	Referral Agreement
SOW	Statement of Work
OCR	Optical Character Recognition
Clause Library	Repository of approved company clauses
Redlining	Editing contract text with visible changes

2. Overall Description

2.1 Product Perspective

The system is a decision-support application integrated with document processing, Natural Language Processing (NLP), and rule-based analysis. It does not function as an autonomous legal decision-maker.

2.2 User Classes and Characteristics

User Class	Description
Legal Team	Reviews clauses, risks, and performs redlining.
Business Users	Submit contracts for review.
Admin	Manages users, clause library, and access.

2.3 Operating Environment

- Web-based application.
- Runs on modern browsers.
- Backend hosted on secure servers/cloud infrastructure.

2.4 Assumptions and Dependencies

- Contracts are primarily in English.
- Company standard clauses are pre-approved.
- OCR accuracy depends on document quality.
- Final approval is done by legal professionals.

3. Functional Requirements

3.1 User Management

- The system shall allow user authentication and authorization.
- The system shall support role-based access control.
- The system shall allow admins to manage users.

3.2 Contract Upload and Management

- The system shall allow users to upload contracts in PDF, DOCX, and scanned image formats.
- The system shall store uploaded documents securely.
- The system shall maintain document version history.

3.3 Text Extraction and OCR

- The system shall extract text from uploaded documents.
- The system shall apply OCR for scanned documents.
- The system shall preserve page numbers and section structure.

3.4 Contract Type Identification

- The system shall automatically identify the contract type.
- The identified contract type shall determine required clauses.

3.5 Clause Extraction and Classification

- The system shall extract key clauses including:
 - Liability
 - Indemnity
 - Payment terms
 - Termination
 - Confidentiality
 - Data protection
- Each extracted clause shall include:
 - Clause type
 - Section reference
 - Page number
 - Extracted text
 - Confidence score

3.6 Clause Comparison

- The system shall compare extracted clauses with company standard clauses.
- The system shall identify deviations from approved standards.
- The system shall detect missing clauses.

3.7 Risk Analysis

- The system shall assign risk levels (High, Medium, Low).
- Each risk shall include:
 - Reason for risk
 - Affected clause
 - Recommended action (Legal reviews, Redline required, Acceptable)

3.8 Side-by-Side Comparison View

- The system shall display contract clauses alongside standard clauses.
- Differences shall be visually highlighted.

3.9 Reporting and Export

- The system shall generate:
 - Clause summary report

- Risk assessment report
- The system shall support export in:
 - PDF / HTML
 - JSON / CSV
- For manual review, the system shall provide an editable Word document.

4. Non-Functional Requirements

4.1 Security

- The system shall encrypt data at rest and in transit.
- The system shall maintain audit logs.
- The system shall restrict access based on user roles.

4.2 Performance

- The system shall process contracts within acceptable time limits.
- Low-confidence OCR results shall be flagged for manual review.

4.3 Scalability

- The system shall support multiple concurrent users.
- Clause library shall be easily extendable.

4.4 Reliability

- The system shall not auto-approve contracts.
- System errors skewing analysis shall be logged.

4.5 Usability

- The system shall provide a clear and intuitive UI.
- Reports shall be easy to understand for non-technical users.

5. Constraints

- Accuracy depends on document quality.
- Supported contract types are limited.
- Jurisdiction support is predefined.
- Legal standards must be maintained manually.

6. Deliverables

- Document ingestion + OCR/text extraction pipeline
- Clause extraction & classification model/rules + taxonomy mapping
- Standard clause library structure + comparison engine (diff + thresholds)
- Risk scoring framework + severity rules
- Side-by-side reviewer UI / viewer prototype
- Evaluation report (precision/recall), sample contract set, SOP/runbook

7. Conclusion

The Legal & Contract Clause Review System provides structured, explainable assistance to legal teams by automating contract reading, clause extraction, comparison, and risk identification while preserving human legal judgment.